Ze-Yu Xing

PhD Candidate in Physics
Institute of Physics, Chinese Academy of Sciences

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Research Profile

My research explores novel phenomena in quantum systems far from equilibrium, particularly in open quantum systems and quantum many-body dynamics. I am interested in the principles governing collective behavior and relaxation, and in how information theory and probability can deepen our understanding of these phenomena. Looking ahead, I aim to explore how AI can advance theoretical discovery and how insights from physics can contribute to quantum computing and new approaches to learning and computation.

Education and Affiliation

Institute of Physics, Chinese Academy of Sciences Aug. 2022 - Present
PhD Candidate in Physics *Beijing, China*
Jilin University Aug. 2018 - Jun. 2022
B.Sc. in Physics, Honors Program in Science *Jilin, China*
GPA: 3.88/4.00 (93.2/100).

Research Interests

Non-equilibrium quantum dynamics · Open quantum systems · Quantum many-body physics · Non-Hermitian physics · Statistical mechanics · Quantum information and simulation

Ongoing Research Projects

OYA-OQS: Liouvillian Lieb-Schultz-Mattis Constraints in Open Quantum Systems
Local, translationally invariant Lindbladians with strong U(1) symmetry and a unique steady state in a fixed-charge sector exhibit asymptotically gapless modes in the thermodynamic limit: long-wavelength modes at any filling, plus additional modes at wave vectors set by non-integer filling.
Strong Quantum Mpemba Effect via Spontaneous Symmetry Breaking in Open Systems
Spontaneous symmetry breaking enables the strong quantum Mpemba effect for discrete or continuous, strong or weak symmetries without delicate state engineering. States in the steady-state symmetry sector bypass slow modes and relax faster than closer states overlapping the slow manifold.

Publications and Preprints

Geometry-adaptive formulation of non-Bloch bands in arbitrary dimensions and spectral instability 2026
Ze-Yu Xing, Yuncheng Xiong, and Haiping Hu
Communications Physics 9, 127 (2026) · arXiv:2407.01296 (2024)
Ze-Yu Xing and Yuncheng Xiong contributed equally.

Universal spreading dynamics in quasiperiodic non-Hermitian systems 2025
Ze-Yu Xing, Shu Chen, and Haiping Hu
Physical Review B 111, L180203 (2025)

Talks and Presentations

Symmetry-Enforced Strong Quantum Mpemba Effect 2025
19th Symposium for Young Scholars on Cold Atom Physics, Changsha, China.

Honors and Awards

Director's Scholarship, Institute of Physics, Chinese Academy of Sciences 2025
Ny Tsi-Ze Scholarship, Institute of Physics, Chinese Academy of Sciences 2020
Honorable Mention, Mathematical Contest in Modeling, Consortium for Mathematics and Its Applications 2020
National Scholarship, Ministry of Education, People's Republic of China 2019
Outstanding Student of Jilin University, Jilin University 2019